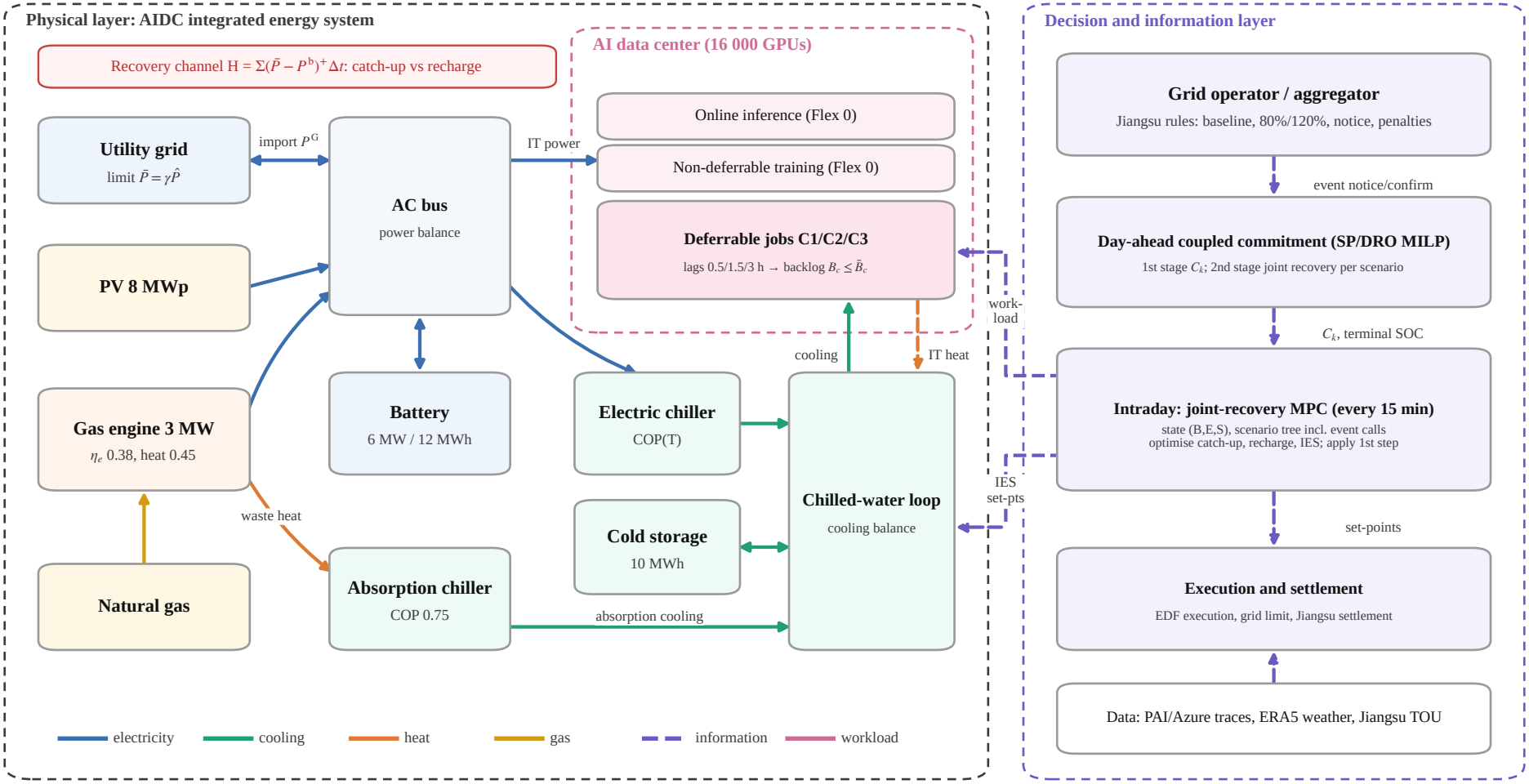
Physical layer: AIDC integrated energy system
Recovery channel H = Σ(P̄ − P^b)^+ Δt: catch-up vs recharge
AI data center (16 000 GPUs)
Online inference (Flex 0)
Non-deferrable training (Flex 0)
Deferrable jobs C1/C2/C3
lags 0.5/1.5/3 h → backlog B_c ≤ B̄_c
Utility grid
limit P̄ = γP̂
import P^G
AC bus
power balance
IT power
PV 8 MWp
Gas engine 3 MW
η_e 0.38, heat 0.45
Natural gas
waste heat
Battery
6 MW / 12 MWh
Electric chiller
COP(T)
cooling
IT heat
Chilled-water loop
cooling balance
Cold storage
10 MWh
Absorption chiller
COP 0.75
absorption cooling
work-load
IES set-pts
Decision and information layer
Grid operator / aggregator
Jiangsu rules: baseline, 80%/120%, notice, penalties
event notice/confirm
Day-ahead coupled commitment (SP/DRO MILP)
1st stage C_k; 2nd stage joint recovery per scenario
C_k, terminal SOC
Intraday: joint-recovery MPC (every 15 min)
state (B,E,S), scenario tree incl. event calls
optimise catch-up, recharge, IES; apply 1st step
set-points
Execution and settlement
EDF execution, grid limit, Jiangsu settlement
Data: PAI/Azure traces, ERA5 weather, Jiangsu TOU
electricity
cooling
heat
gas
information
workload